Pintoo Kumar

Senior MERN Stack Engineer | Node.js, React.js, Next.js, MongoDB, Docker, Kubernetes

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Professional Summary

Full Stack Engineer with 8+ years of experience designing and developing scalable MERN applications and microservices. Expert in building REST and GraphQL APIs using Node.js (Express, NestJS) and modern UIs with React.js and Next.js. Strong in architecting cloud-native applications using Docker, Kubernetes, and CI/CD pipelines on AWS and Azure.

Proficient in integrating AI-powered automation (OpenAI, Claude, Cursor) and real-time capabilities using WebSockets and Supabase Realtime. Skilled at database design and optimization with MongoDB and PostgreSQL, delivering high-performance and maintainable systems using clean code and system design best practices.

Core Skills

- **Frontend:** React.js, Next.js, Redux, MUI, TailwindCSS, TypeScript
- **Backend:** Node.js, Express.js, NestJS, GraphQL (Apollo Server/Client), REST, WebSockets
- **Database:** MongoDB, PostgreSQL, Redis, Supabase (Auth, Edge Functions, Realtime)
- **DevOps & Cloud:** Docker, Kubernetes, Helm, GitHub Actions, Jenkins, AWS, Azure, DigitalOcean
- **AI & Automation:** OpenAI GPT-4o, Claude 3, v0.dev, Cursor, LLM-powered workflows
- **Testing & Quality:** Jest, Cypress, Mocha, Chai, Prometheus, Grafana
- **Other:** Supabase Auth, Edge Functions, CI/CD, Agile, Swagger

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Selected Projects

Supply Chain Management System

A blockchain-based vaccine shipment tracker system that monitored real-time temperature and geolocation during transit.

Architected and implemented an end-to-end **MERN-based e-commerce system** with advanced inventory, payments, and CRM modules.

- Built backend services with **Node.js (Express, GraphQL)** and **MongoDB** for flexible schema management.
- Integrated **Stripe and Razorpay** for payment handling and developed real-time order tracking.
- Designed responsive **Next.js + MUI** frontend with Redux-based state management.
- Deployed containerized microservices using **Docker & Kubernetes** with CI/CD pipelines.
- Implemented role-based authentication and monitoring via **Prometheus & Grafana**.

DeFi Lending Platform

A blockchain-powered decentralized finance platform enabling tokenized lending and collateralized borrowing.

- Architected secure blockchain interactions via **Hyperledger Fabric** and implemented business logic in **Golang** smart contracts.
- Set up decentralized network orchestration using **Shell scripting**, **Docker containers**, and **CouchDB** for on-chain persistence.
- Developed backend services with **Node.js**, RESTful APIs, and leveraged **MongoDB** for fast off-chain record management.
- Built responsive, single-page UI components using **React.js**, **Redux**, **TailwindCSS**, and **TypeScript**, connecting via **Axios** to blockchain APIs.
- Ensured consistent application quality through automated testing in **Jest** and Git-managed CI workflows.
- Added AI-powered automation using OpenAI GPT-4o and Claude 3.
- Implemented Supabase Edge Functions for event-driven workflows.

E-Commerce Platform

- Architected backend with **Golang (Gin, GORM)** and GraphQL APIs for inventory, payments, and CRM modules.
- Integrated Stripe and Razorpay; optimized PostgreSQL queries for high concurrency.
- Deployed with Docker/Kubernetes and implemented CI/CD pipelines.

Web3 Backend APIs
Backend services for a decentralized **NFT marketplace** and **token issuance platform**.

- Developed REST and GraphQL APIs in **Node.js/NestJS** for blockchain data interaction.
- Integrated **ethers.js** for EVM contract operations and wallet management.
- Used **MongoDB + Redis** for transaction and caching layers.
- Wrote **unit and integration tests** using Jest and managed Swagger documentation.

Title-Direct
A real estate and mortgage platform for title management and advisory services.

A **real estate and mortgage management** system built on a scalable MERN + DevOps stack.

- Built secure APIs with **NestJS (GraphQL)** and **MongoDB** using Prisma ORM.
- Integrated **Azure AD authentication**, blockchain ledger (R3 Corda), and Redis caching.
- Deployed services via **Docker, Helm, and Azure AKS**; monitored with **New Relic**.
- Built an admin dashboard using **Next.js and MUI** for workflow visualization.

DocVenture
A digital document automation platform for secure contract generation and approvals.

A digital document automation platform for **secure contract creation and AI-assisted approvals**.

- Developed full-stack components with **NestJS (GraphQL)**, **MongoDB**, and **Prisma**.
- Added **AI-powered automation** using OpenAI GPT-4o and Claude 3 for document generation.
- Designed frontend with **Next.js, MUI, and Apollo Client** for real-time collaboration.
- Deployed on **Azure** with GitHub Actions–based CI/CD and monitored with **Grafana**.

Education & Certifications

- B.Tech in Electrical Engineering – Krishna Institute of Engineering & Technology
- **Docker & Kubernetes: The Complete Guide** – Udemy
- **Full Stack Web Development with Node.js and React** – Udemy